
FLOWER COLLABORATIVE AGENT HACKATHON · CAMBRIDGE 2026

Provenance Harness

A chain of custody for machine reasoning.

Every step an AI takes, sealed like courtroom evidence — checkable by anyone, rewriteable by no one.

MUNIB RAHMAN · SEEVAD · TRACK 1: SUPERGRID

WHO LIVES WITH THE PROBLEM

The most valuable questions are locked in separate rooms.

The data exists. The permission to assemble it does not.

Across whole industries, the data needed to answer one question is split between organisations that cannot pool it — by law, or by rivalry.



Healthcare

One patient's scans, genome and trial history sit in three different trusts; none may let its page leave the building.



Banking

A fraud ring only becomes visible across several banks — yet no bank may pool customer records with its rivals.



Pharma

Trial results stay siloed by company and site, so replication crawls — and so does every cure waiting on it.



Public services

Housing, benefits and care live in separate agencies; a caseworker only ever sees a fragment of a person.



Science

Published claims rest on unpublished data that other labs cannot inspect.



Energy

Grid operators cannot share live telemetry across competitors — even when the stability of the grid depends on it.

WHO WINS WHEN IT IS SOLVED

The same rooms — finally allowed to answer.

When the question travels instead of the data, shelved questions come back to life — for the institutions, and for everyone who must trust their answers.

Hospitals → Diagnoses that cross trust boundaries, with a trail a clinician can defend.

Banks & regulators → Cross-bank fraud and money-laundering cases, with evidence a court accepts.

Insurers → Claims settled against a record no party can quietly dispute.

Pharma → Multi-site trials without pooling a single raw patient record.

Scientists → Results verified without the data ever being surrendered.

AI builders → Systems that can finally be sold into regulated industries.

The data never moves. Only the question does.

ONE QUESTION, THREE ROOMS

One patient. Three hospitals. No way to ask.

Amira is 54. Her scans live in one hospital trust, her genomic record in a second, her trial history in a third. Each page may never leave its room — the specialists can call facts through the wall, but no page can pass under the door.

*“Did the treatment work for patients like her?”
Today, that question simply goes unanswered.*

Not because the reasoning is hard — because the evidence cannot be assembled, and nobody could audit the answer if it were.

This is the motivating case, not what the demo runs on — the demo attests two data shards.



TRUST A — SCANS · TRUST B — GENOME · TRUST C — TRIAL HISTORY



The reasoning evaporates.

A multi-agent system thinks out loud, scattered across machines. Today that thinking lands in log files — and a log file was never evidence.

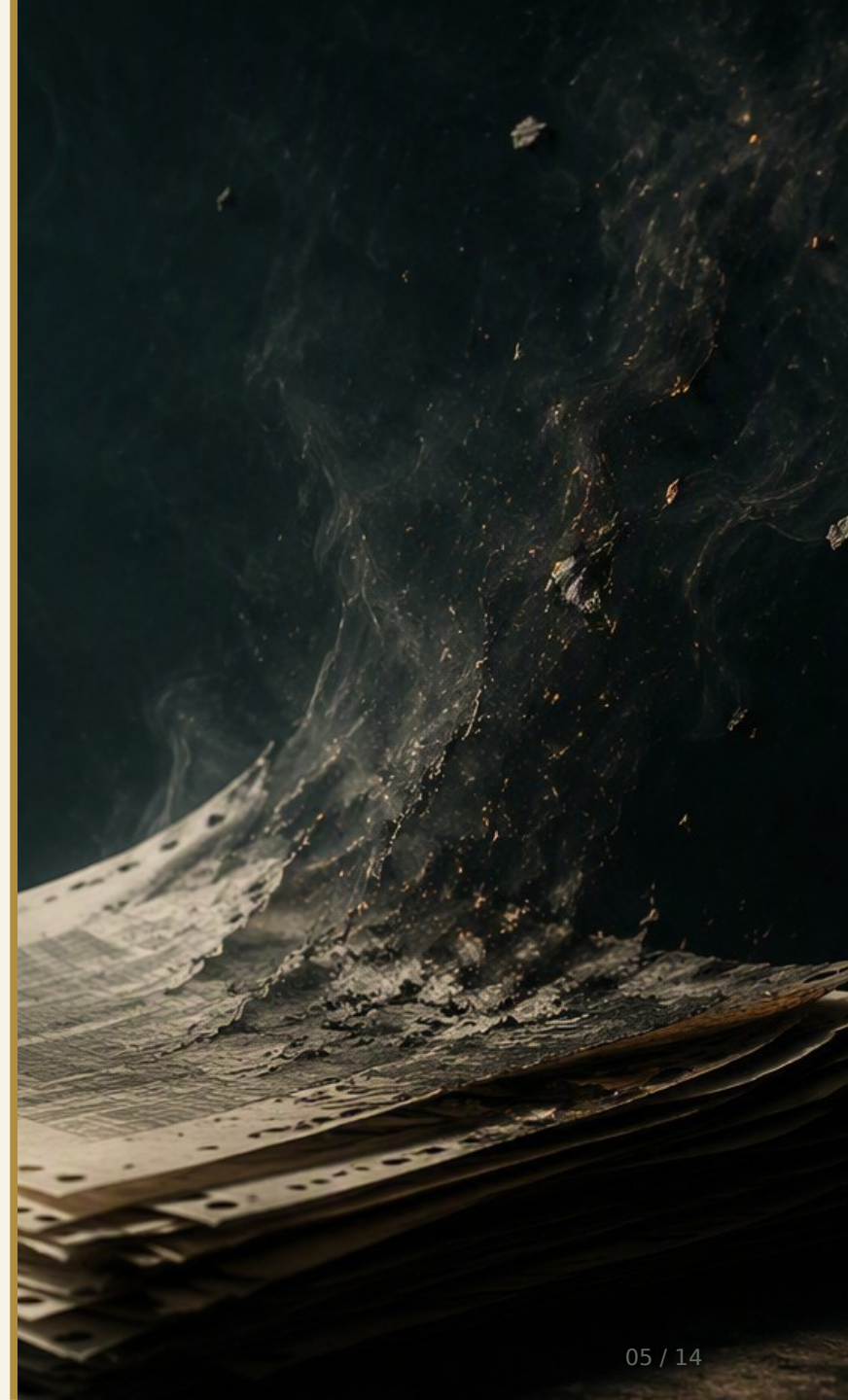
Logs can be edited ...and nothing inside the file will ever tell you it happened.

Logs can be lost ...a rotated file, a crashed container, a cleared buffer.

Logs can be reordered ...a timestamp is only a number someone typed.

*“A regulator never asks ‘what did the system say?’
They ask: show me how it got there — and prove
nobody touched the record.”*

In health, finance and public services, a conclusion without a defensible trail is worth nothing at all.





WHAT COURTS ALREADY KNOW

Evidence without custody is just a story.

From crime scene to courtroom, every hand that touches the evidence signs the bag. Every seal stays unbroken. Break the chain once — and the truth itself becomes inadmissible, even when it is real.

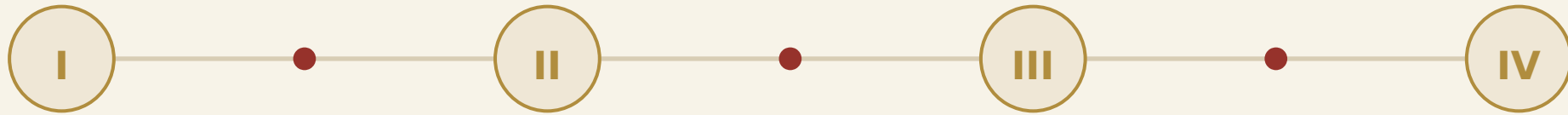
AI answers travel with no bag, no seal, no signature

We built one.

WHAT WE BUILT

A harness, not an app.

A harness is the scaffolding agents run inside — the seatbelts and the flight recorder, not the destination. Any multi-agent task dropped into it inherits the same four custody rules.



Question separately

Each holder answers on its own. Like witnesses, they never confer — nothing is merged into one pile.

Seal each statement

Every contribution comes back with a fingerprint of the answer — never the records themselves.

Chain the seals

Each entry's seal is pressed from the seal before it, all the way back to entry zero.

Sign off, or refuse

Conditions unmet? The round is not signed. Mechanically.

Every claim carries a fingerprint. Every entry seals the one before it.

Nothing is signed off unless the conditions are met — and anyone can check the whole chain, without trusting us.

The manifest, not the crate.

Customs never opens every container. It reasons from a sealed manifest — weight, origin, category, seal number — while the cargo stays locked inside. That is exactly what crosses the wall here.

Each holder's agent receives a fingerprint of the contribution — a declaration about the records, never the records themselves.

MANIFEST — HOLDER B

hash 9f2e...c41a

weight 0.87

origin genomic-panel

cargo never leaves

*Asked precisely? Say it
precisely — the analogy is a
door into the idea, not a
substitute for the truth.*

This is an integrity property, not a confidentiality one. The digest is computed locally and never taken on trust from the model. Real confidentiality would need attestation running as a ClientApp on each node — it doesn't yet, and the README says so.

A chain of wax seals.

Each page of the ledger is sealed in wax — and the stamp that presses each seal is carved from the seal on the page before it. To change one page quietly, you must re-forge every seal that follows. Each one depends on the last.



intact — each seal matches the one before

someone edited entry 5 — every seal after it fails

“Edit anything in the middle, and every seal after it breaks. You cannot quietly rewrite this record.”

Say this one out loud — it is the whole mechanism in a single sentence.

Two keys, or the door stays shut.

Some doors need two keys turned at once, by two different people — not because anyone is suspected, but because a single point of failure is unacceptable for what sits behind the door. The harness will not sign a round off as reproducible while a holder is missing.

A model that decides to be cautious is a hope. A check in the harness is a property.

If a judge asks which one this is, the answer is the second — and you can show the code.

Two distinct attestations required. Two contributions with the same digest are refused — duplicate content cannot stand in for independent work. Recorded as `policy.satisfied` or `policy.refused`.



Holder A —
turned



Holder B —
missing



NOT SIGNED OFF

the round is refused, mechanically

decided by: `harness policy` (the model does not get a vote)

The model's text is printed separately under **MODEL COMMENTARY** (not the decision).

Hold it up to the light.

You don't trust a £20 note because the person handing it over seems honest — you hold it to the light and check the watermark yourself. The verifier is that light: a standalone command that takes only the ledger file. No access to my machine. No trust in me.

```
VERIFY — STANDALONE, LEDGER FILE ONLY
```

```
$ ./demo.sh
```

```
VERIFIED — 10 entries, chain intact
```

```
$ ./break.sh
```

```
changed ONE character inside a hash:
```

```
3afe386929f903f46e39...
```

```
-> 3afe386929b903f46e39...
```

```
LEDGER BROKEN
```

```
first bad entry : #5 (kind=tool.call actor=harness)
```

```
TRUSTWORTHY : entries 0..4 (5 of 10)
```

```
NOT TRUSTWORTHY : entries 5..9 (5 of 10)
```

Two commands. I run the first, you run the second.

Then hand over the keyboard: you give a stranger the means to attack your system, in public — and it holds.

THE LIMIT — SAID BEFORE YOU ASK

A sealed bag can still hold planted evidence.

Forensics can prove the bag was never opened between the scene and the courtroom. It cannot prove what went into the bag was real. A hash chain has the same blind spot — a chain of lies verifies perfectly.

WHAT WOULD CLOSE IT — I

Signed contributions

Each holder signs its own contribution with its own key — so no single writer can fabricate another holder's claim.

WHAT WOULD CLOSE IT — II

External anchoring

The head hash is anchored outside the system — so the chain cannot be rewritten wholesale and backdated.

Founders and academics are trained to hunt the overclaim. Hand it over unprompted — and everything else you said gets believed.

WHERE IT GOES

One laptop today. Thirty hospitals unchanged.

The clinical case is the demonstration, not the product. The harness is domain-agnostic — it goes anywhere agents must reason across organisations that cannot pool their data.

Health

Genomic and clinical records that cannot leave a trust — but a diagnosis that must be defensible.

Finance

Patterns visible only across banks, delivered with an evidence pack a regulator will accept.

Public services

Housing, benefits and care joined up for a caseworker, without merging the databases.

Science

Claims checked against unpublished data held at other labs — with the check itself on record.

RUNNING TODAY

Flower AgentApp · local SuperLink · ledger emitted via agent.events.emit() · Qwen3.5 397B on AMD Instinct MI300X
published on Flower Hub as @munibrahman/provenance-harness · open source



IF THEY REMEMBER ONE SENTENCE

**You cannot quietly
rewrite this record.**

And you don't have to take my word for it — check it yourself.

github.com/munibrahman-star/provenance-harness
Munib Rahman · SEEVAD · munibrahman@gmail.com